

Reflection on the development of student dropout prediction project

Introduction

In developing a dropout prediction model for our institution, we learned that high accuracy does not equal high utility. Our goal was to build a safety net to identify at risk students before they leave. However, because this system influences human intervention determining who receives counseling and financial aid the ethical stakes are high. A False Negative is not just a statistical error, it is a student falling through the cracks. This reflection outlines the risks inherent in our pipeline and the safeguards required to deploy this system responsibly.

Risk Analysis

We found considerable hidden biases in our historical data, since the code learns from this data we encounter a few risks

- Proxy Bias: We intentionally removed gender and age to prevent direct discrimination. However, the model found a proxy: Tuition Debt. Since debtor status is a strong predictor of dropout, and Scholarship students rarely have debt, the model incorrectly assumed Scholarship students were "Safe." This resulted in a Recall rate of only 54% for this vulnerable group compared to 90% for the general population. However, without a more extensive historical dataset and rigorous causal analysis there could be additional proxies.
- Data Scarcity: International students comprise less than 3% of our data. The model struggled to learn their specific risk patterns, treating them as statistical noise and leading to a system that works well for locals but fails the minority.
- If a student is informed that an AI system has flagged them as "High Risk," it can create a devastating "Labeling Effect." The student may internalize this prediction, lose confidence and ironically cause the very dropout the system was meant to prevent.
- Automation bias: the tendency for counselors to trust the machine over their own intuition. If the model flags a student as high risk a counselor might unconsciously treat them as a lost cause.

Mitigation and safeguards

To fix the Scholarship gap, we applied Inverse Class Reweighting specifically to that subgroup. This forced the model to look beyond financial signals and find other warning signs (like grades or attendance). This intervention raised the Recall for scholarship holders from 54% to 67%, closing the safety gap without distorting the model for others.

For International students, algorithmic fixing (over weighting) would have introduced instability and overfitting due to the small sample size. Instead, we implemented a governance safeguard International students are automatically routed to a human counselor check in, bypassing the AI risk score entirely. This acknowledges that AI is not the right tool for every demographic.

To mitigate the labelling effect somewhat we modified the counselor protocol, when AI flags a student, the counselor initiates a generic “check-in” framing it as a routine outreach conversation. In addition, based on our audit session we have realised that an exit strategy should be clearly defined and should not be something that we say we will do. In that case, the principle would be formally checked but would lack any meaningful enforcement. The same applies to monitoring the algorithm. Regular monitoring sessions, held monthly, should be conducted to evaluate the model’s performance in collaboration with student counselors. These sessions would allow for timely adjustments, such as revising variables or conducting additional research to determine which variables are most appropriate, based on students feedback. More data is needed to generalize better.

Opportunities

By using an interpretable model (Logistic Regression), we can guide counselors toward the likely type of support needed (e.g., Financial Aid vs. Academic Tutoring), making the limited counseling resources more efficient. Additionally, in response to feedback on our algorithm, more resources should be allocated to identifying additional variables relevant to student performance, for example through targeted questionnaires and structured input from counselors.

Conclusion

Should we deploy? Yes, but only under strict conditions.

Given the limitations in our proxy analysis, we cannot yet claim the model is fully “fair” in a causal sense, and the limited context makes for an algorithm that needs a lot of additional work. Therefore, the system should be deployed as a Decision Support Tool, not a Decision Maker.

No Automated Penalties: The AI cannot be used to restrict access to courses or remove privileges.

Human Firewall: A counselor must verify every flag before acting.

Based on the feedback from the presentation, we argue that if the risk score is never disclosed to students, thereby preventing stigma, and if human oversight is used to compensate for the model’s blind spots, the system can serve as a valuable support tool. Under these conditions, it helps identify and assist students who might otherwise fall through the cracks..